THADOMAL SHAHANI ENGINEERING COLLEGE

DEPARTMENT OF

Artificial Intelligence & Data Science

Name of Student : Adit Sanjay Mokashi

Class : SE

Division : S2

Roll Number : 2511136

Batch : S23

Subject : Fundamentals of Telecommunication **Systems**

Experiment Number : 1

Experiment Title : Study and Operation of a Basic Telephone System

Date of Submission : 02/04/2026

Experiment No. 01

Title: Study and Operation of a Basic Telephone System

Aim:

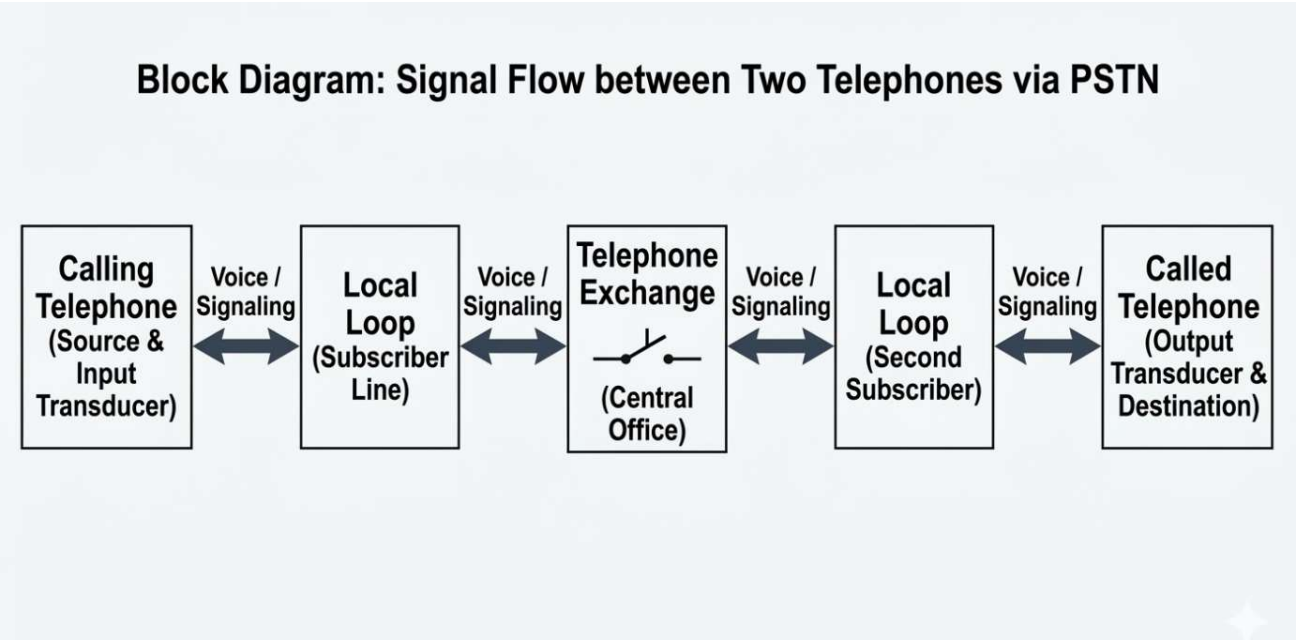
To study and understand the construction, components, and operation of a basic telephone system using online resources and videos.

Theory

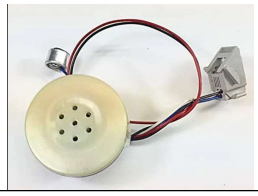
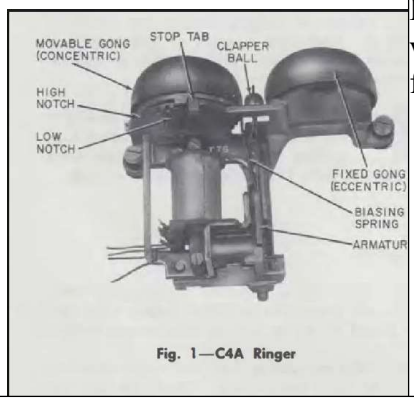
A basic telephone system enables voice communication between two distant users by converting speech signals into electrical signals, transmitting them through a communication network, and reconverting them back into sound at the receiver.

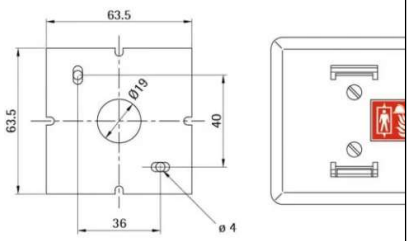
Traditional telephony operates on the Public Switched Telephone Network (PSTN), which uses circuit switching to establish a dedicated communication path for the duration of a call.

Block Diagram



Task 1: The "Inside the Handset" Investigation

Component	Visual Reference (Paste Screenshot/Sketch Here)	Technical Function
Microphone		Converts the speaker's voice (sound waves) into electrical signals that can be transmitted through the telephone circuit.
Speaker		Converts incoming electrical voice signals back into sound waves so the user can hear the caller.
Switch Hook		A mechanical switch that detects whether the handset is lifted or placed back, signaling the system to start or end a call.
Ringer		Produces an audible alert (ringing sound) when an incoming call signal is received from the exchange.

Keypad		Allows the user to dial numbers and input commands by generating specific electrical signals (DTMF tones) for each key pressed.
Exchange		The central switching system that connects the handset to other telephones, routes calls, and manages communication signals.

Task 2: Basic Blocks of a Telephone System

Instructions:

Write 3–4 technical sentences describing the function and importance of each block listed below.

1. **Microphone (Transmitter):** The microphone (transmitter) converts acoustic sound waves from the speaker's voice into corresponding electrical signals using a transduction mechanism (e.g., carbon or electret element). Variations in air pressure cause changes in electrical current or voltage, producing an analog representation of speech. This signal is then amplified and transmitted through the local loop to the exchange. The microphone is critical for ensuring accurate capture of voice frequencies for clear communication.

2. **Local Loop:** The local loop is the physical communication link, typically made of twisted-pair copper wires or fiber, connecting the subscriber's telephone to the nearest telephone exchange. It carries bidirectional analog or digital voice signals between the user and the telecom infrastructure. The quality and bandwidth of the local loop directly affect call clarity and signal strength. It is often referred to as the "last mile" in telecommunication systems.

3. **Telephone Exchange:** The telephone exchange is a central switching system that establishes, maintains, and terminates connections between multiple subscribers. It processes dialing information, determines the destination of the call, and routes signals through appropriate communication paths. Modern exchanges use digital switching techniques for efficient traffic management and signal processing. The exchange plays a crucial role in enabling large-scale, reliable voice communication networks.

4. **Transmission Network:** The transmission network consists of high-capacity communication links such as fiber optic cables, microwave systems, or satellites that interconnect multiple exchanges. It carries voice signals over long distances with minimal attenuation and distortion. Advanced multiplexing and digital encoding techniques improve efficiency and reliability of signal transmission. This network forms the backbone of modern telecommunication systems.

5. **Receiver (Speaker):** The receiver converts incoming electrical signals back into audible sound waves using electromagnetic or dynamic speaker principles. Electrical variations cause a diaphragm to vibrate, reproducing the original speech signal. The receiver must maintain clarity and minimal distortion across voice frequency ranges. It ensures that the listener accurately perceives the transmitted message.

6. **Ringer and Dialing Circuit:** The ringer circuit generates an audible alert when an incoming call signal is received from the exchange, typically using an electromechanical bell or electronic tone generator. The dialing circuit allows the user to input numbers via keypad or rotary dial, producing DTMF (Dual Tone Multi-Frequency) signals corresponding to each digit. These signals are interpreted by the exchange to route the call to the intended recipient. Together, these circuits enable call initiation and notification functionalities.

Task 3: Conceptual Working of a Telephone System

Instructions:

Based on virtual demonstrations, write 2 to 3 sentences explaining each step.

1. Off-Hook Condition: When the user lifts the handset, the switch hook mechanism closes the circuit, signaling the telephone exchange that the line is active. This action allows current to flow through the local loop, indicating readiness to initiate a call. The system then prepares to provide a dial tone.

2. Reception of Dial Tone: After detecting the off-hook condition, the exchange sends a continuous dial tone through the line. This tone informs the user that the connection to the exchange has been successfully established. It also indicates that the system is ready to receive dialing input.

3. Dialing (DTMF Generation): When a user presses keys on the keypad, each button generates a unique combination of two frequencies known as Dual Tone Multi-Frequency (DTMF) signals. These signals travel through the local loop to the telephone exchange. The exchange decodes the tones to identify the dialed number.

4. Switching & Routing: The telephone exchange processes the received digits and determines the appropriate path to connect the call to the destination number. Using switching systems, the exchange establishes a temporary communication circuit between the caller and receiver. This process ensures efficient routing through the transmission network.

5. Ringing State (Ring-back vs. Ringing): Once routing is complete, the exchange sends a ringing signal to the called party's telephone, activating its ringer. Simultaneously, the caller hears a ring-back tone indicating that the destination phone is being alerted. This stage confirms that the connection attempt is in progress.

6. Voice Transmission (The Talking State): When the called party answers the phone (off-hook), a complete communication path is established between both users. The microphone converts speech into electrical signals, which are transmitted through the network in real time. The receiver converts the electrical signals back into sound, enabling two-way conversation.

7. On-Hook (Call Termination): When either user places the handset back on the cradle, the switch hook opens the circuit, stopping current flow. The exchange detects the on-hook condition and disconnects the communication path. All network resources allocated for the call are then released for future use.

Conclusion

From the virtual demonstrations, we understood that a telephone system converts human voice into electrical signals using a microphone. These signals travel through the local loop to the telephone exchange, where switching systems determine the correct destination and establish a communication path through the transmission network. The electrical signals carry the characteristics of the speaker's voice in real time across long distances. At the receiver end, the speaker converts these electrical signals back into sound waves, reproducing the original speech. The complete process demonstrates how multiple components and network infrastructure work together to enable reliable long-distance voice communication.